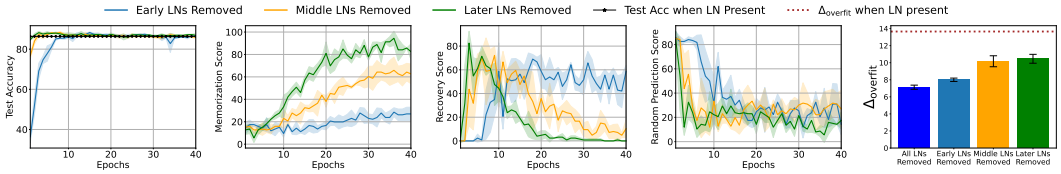


Figure 24: **Pivotal impact of early LNs on memorization in Post-LN model (ELECTRA, News Dataset).** We can clearly observe the impact of early layers LN removal on suppressing memorization & achieving true label recovery in ELECTRA, accompanied with higher train-test-accuracy gap, $\Delta_{\text{overfit}}^{\text{Pre, early}}$, than for later layers, $\Delta_{\text{overfit}}^{\text{Pre, later}}$, while learning being intact.

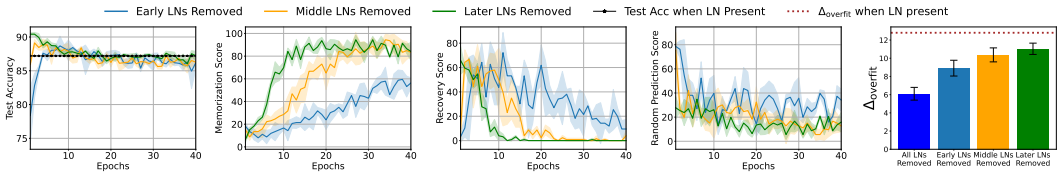


Figure 25: **Pivotal impact of early LNs on memorization in Post-LN model (Longformer, News Dataset).** We can clearly observe the impact of early layers LN removal on suppressing memorization & achieving true label recovery in Longformer, accompanied with higher train-test-accuracy gap, $\Delta_{\text{overfit}}^{\text{Pre, early}}$, than for later layers, $\Delta_{\text{overfit}}^{\text{Pre, later}}$, while learning being intact.

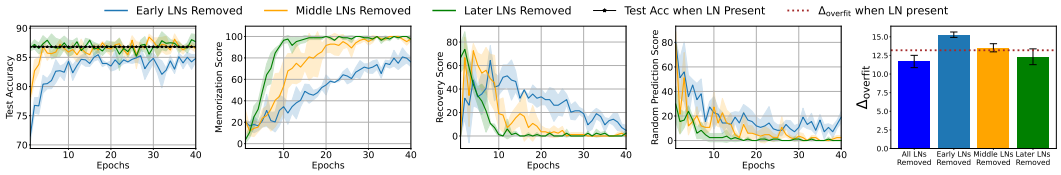


Figure 26: **Pivotal impact of early LNs on memorization in Post-LN model (RoBERTa, Tweet-Topic Dataset).** We can clearly observe the impact of early layers LN removal on suppressing memorization & achieving true label recovery in RoBERTa, while learning being minimally impacted. *Note: In the case of RoBERTa, $\Delta_{overfit}^{Pre,\ early}$ is slightly higher than $\Delta_{overfit}^{Pre,\ early}$, because even though removing early LNs, led to a greater memorization suppression than later LNs, but learning also got affected slightly, hence, the overfitting gap increased. However, we need to compare this Post-LN result with its Pre-LN counterpart of GPT2 (Fig. 19), where early LNs removal, affected learning even more, without suppressing memorization.*

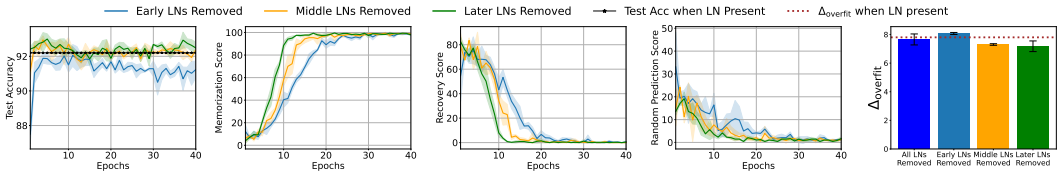


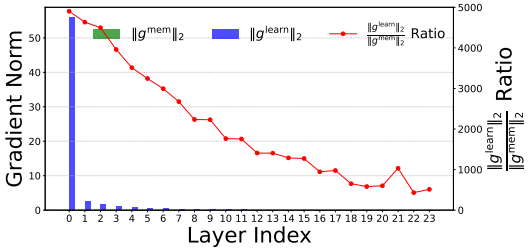
Figure 27: **Pivotal impact of early LNs on memorization in Post-LN model (DistilBERT, Emotions Dataset).** For the case of DistilBERT, we observe that just like when all LNs removal could not suppress memorization, the same happens with early LNs removal. However, we want to draw attention to the delay in achieving memorization, i.e., early LNs removal results in the highest delay in memorization compared to Middle/Later LNs removal. This shows that Early LNs are still the most significant.

G.3 Gradients explain the impact of LN on Memorization & Learning

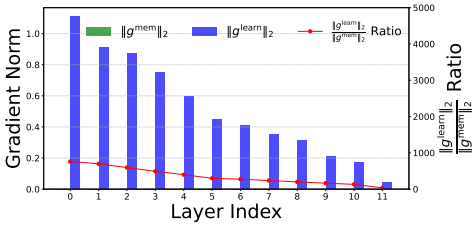
In this section, we provide additional results to understand the distinctive impact of LN on memorization and learning through the lens of gradients. These results provide a deeper understanding of why (1) LN removal destabilizes learning in Pre-LN models, and suppresses memorization in Post-LN models, (2) Early Layers LN are more significant than later layers LN in driving these phenomena.

G.3.1 Language Datasets Results

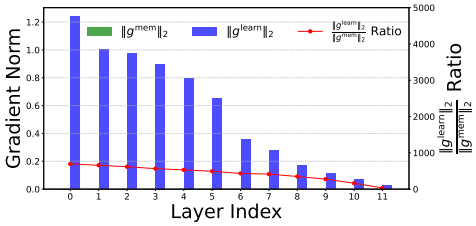
In the language modality, we experimented with additional Pre-LN (GPTNeo, GPT2, Qwen2) and Post-LN models (BERT, RoBERTa, Longformer, ELECTRA) across Emotions, News, and Tweets Datasets. The results for the same are depicted in Figs. 28, 29, and 30.



(a) Pre-LN (Qwen2) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers

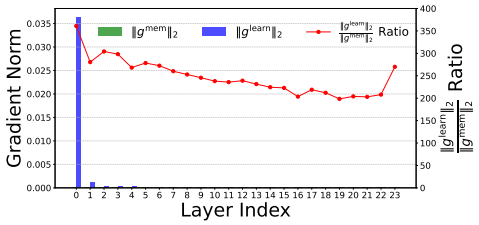


(b) Post-LN (ELECTRA) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers

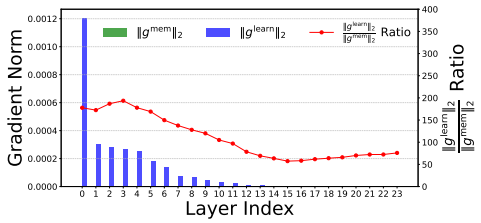


(c) Post-LN (Longformer) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers

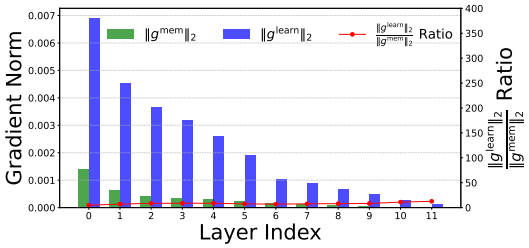
Figure 28: **Learning vs. Memorization Gradients in Pre- and Post-LN Models (News Dataset):** Results clearly exhibit high gradient norms of early layers LNs than later layers for both learning and memorization in Pre-LN (Qwen2) and Post-LN (ELECTRA, Longformer) models. Importantly, the learning gradient norms ($\left\|g_x^{\text{learn}}\right\|_2$) are consistently higher than the memorization gradient norms ($\|g_x^{\text{mem}}\|_2$) across all layers. Furthermore, the ratio $\left\|g_x^{\text{learn}}\right\|_2 / \|g_x^{\text{mem}}\|_2$ is significantly higher in Pre-LN models compared to Post-LN models.



(a) Pre-LN (GPT2) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers

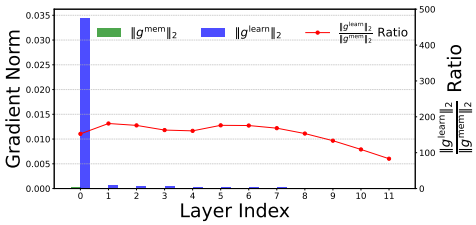


(b) Pre-LN (RoBERTa-PreLayerNorm) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers

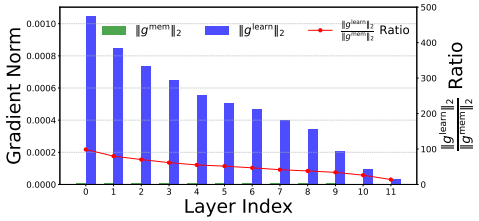


(c) Post-LN (RoBERTa) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers

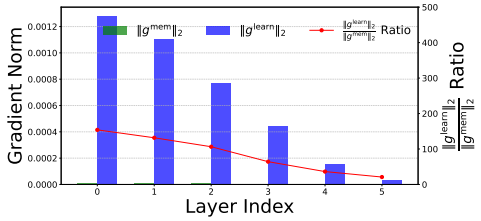
Figure 29: **Learning vs. Memorization Gradients in Pre- and Post-LN Models (TweetTopic Dataset):** Results clearly exhibit high gradient norms of early layers LNs than later layers for both learning and memorization in Pre-LN (GPT2, RoBERTA-PreLayerNorm) and Post-LN (RoBERTa) models. Importantly, the learning gradient norms ($\left\|g_x^{\text{learn}}\right\|_2$) are consistently higher than the memorization gradient norms ($\|g_x^{\text{mem}}\|_2$) across all layers. Furthermore, the ratio $\left\|g_x^{\text{learn}}\right\|_2 / \|g_x^{\text{mem}}\|_2$ is significantly higher in Pre-LN models compared to Post-LN models.



(a) Pre-LN (GPTNeo) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers



(b) Post-LN (BERT) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers



(c) Post-LN (DistilBERT) $\left\|g_x^{\text{learn}}\right\|_2$ & $\|g_x^{\text{mem}}\|_2$ analysis across layers

Figure 30: **Learning vs. Memorization Gradients in Pre- and Post-LN Models (Emotions Dataset):** Results clearly exhibit high gradient norms of early layers LNs than later layers for both learning and memorization in Pre-LN (GPTNeo) and Post-LN (BERT, DistilBERT) models. Importantly, the learning gradient norms ($\left\|g_x^{\text{learn}}\right\|_2$) are consistently higher than the memorization gradient norms ($\|g_x^{\text{mem}}\|_2$) across all layers. Furthermore, the ratio $\left\|g_x^{\text{learn}}\right\|_2 / \|g_x^{\text{mem}}\|_2$ is significantly higher in Pre-LN models compared to Post-LN models.